

ASP.NET Core Web API - 2 Years Experience Cheat Sheet

Core Concepts

- .NET Core is cross-platform, faster, and modular.
- Use appsettings.json for configuration.
- Dependency Injection is built-in.
- Kestrel is the web server for ASP.NET Core.

Web API Essentials

- RESTful APIs follow HTTP methods: GET, POST, PUT, DELETE.
- Use [HttpGet], [HttpPost], etc., for routing.
- Use IActionResult or ActionResult<T> as return types.
- Model binding binds HTTP request data to action parameters.

Entity Framework Core

- Use DbContext to interact with the database.
- Code First with Migrations for schema updates.
- Loading types: Lazy, Eager, Explicit.
- Use LINQ for querying.

Security

- Use JWT for token-based authentication.
- Use [Authorize] for protected routes.
- Authentication validates identity, Authorization checks permissions.

Filters & Middleware

- Filters: Authorization, Action, Result, Exception.
- Middleware handles requests in pipeline (e.g., logging, auth).
- Custom middleware via app.Use().

Common Programming Tasks

- Logging via ILogger<T>.
- CRUD using EF Core.
- Create custom Exception handling middleware.
- Implement Pagination using Skip() and Take().

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Tools & Testing

- Use Postman for testing APIs.
- Swagger/OpenAPI for API documentation.
- Moq for mocking services.
- Use xUnit for writing tests.